Diploma in Information Technology

信息技术文凭

Problem Solving

问题解决

Instruction for CA2 Group Assignment

CA2小组作业说明

Apr 2026 Semester

2026年4月学期

Assessment

评估
125 marks (This assignment constitutes 40% of the overall assessment + 10% class participation)
总分125分（本作业占总成绩40% + 课堂参与10%）

Deliverables

交付成果
There are FIVE (5) deliverables in this assignment, students must complete ALL components.
本作业包含**5项**交付成果，学生必须完成所有模块。

1. Written project report (10 marks)
    1. 项目书面报告（10分）
2. Python codes (50 marks)
    2. Python代码（50分）
3. User guides with sample inputs/outputs (15 marks)
    3. 含输入/输出示例的用户指南（15分）
4. Oral Presentation (25 marks)
    4. 口头汇报（25分）
5. Demonstration of non A.I component (25 marks)
    5. 非AI模块演示（25分）

The Task

作业任务
The objective of this project is to allow students to design and implement a mini program.
本项目目标是让学生设计并实现一个小型程序。
You are to use flowchart to design your program, with clear steps and flows indicated.
需使用流程图设计程序，清晰标注步骤与流程。
You need to use Python programming language to code all the programs, according to the project requirement.
需按照项目要求，使用Python编程语言编写全部程序。

In your report, write a short description, in 500 words, on your program design.
在报告中撰写一段约500词的程序设计简要说明。
You need to include the flowchart of your mini program in the report.
报告中需包含小程序的流程图。
You will also include screenshot samples on how the program should run in the report.
报告中还需包含程序运行效果的示例截图。

You will be allowed to use SIM approved Artificial Intelligence tools (Visual Studio Code + Copilot) for coding.
允许使用学院认可的AI工具（Visual Studio Code + Copilot）辅助编码。
However, you are to manually code or improve 1 feature of the project.
但必须手动编写或优化项目中的**1个功能**。

In your Python codes, include comments to explain the purpose of each section.
Python代码中需添加注释，说明每部分代码的作用。

Assessment Marks Allocation

评分分值分配

Component Assessed	Marks Allocation
Part 1 – Report with flowchart	10
Part 2a – Main Programs (Python)	40
Part 2b – Additional Features	10
Part 3 – User guides with screenshots	15
Part 4 – Oral Presentation	25
Part 5 – Demonstration of non-AI component	25
Total	**125**

中文对照

评估模块	分值分配
第一部分 - 含流程图的报告	10
第二部分a - 主程序（Python）	40
第二部分b - 附加功能	10
第三部分 - 含截图的用户指南	15
第四部分 - 口头汇报	25
第五部分 - 非AI模块演示	25
总分	**125**

The Case

项目案例
Mr Arnold is an entrepreneur who owns a fitness studio.
阿诺德先生是一位企业家，经营着一家健身工作室。
He is looking to capitalize on the trend of people wanting to exercise and keep fit, and he wants to create a system that will allow him to easily manage trainer schedules and bookings.
他希望借助大众健身热潮，搭建一套系统，方便管理教练日程与课程预约。
The system should also be able to allow him to analyze and predict how many classes and trainers he needs to hire.
该系统还需支持分析与预测所需开设的课程数量及需聘用的教练人数。

Mr Arnold's studio has 4 rooms – 2 big rooms that can hold 20 people, and 2 small rooms that can hold 10 people, allowing 4 classes to be conducted at the same time.
阿诺德先生的工作室有4间教室：2间大教室可容纳20人，2间小教室可容纳10人，可同时开展4节课程。
He plans to offer 5 different types of classes – Pilates, Yoga, Mixed Martial Arts (MMA), Boxing and KPop Fitness.
他计划开设5类课程：普拉提、瑜伽、综合格斗（MMA）、拳击、韩流健身。

Mr Arnold has a pool of part-time trainers that can teach various classes, but they are not free all the time.
阿诺德先生有一批兼职教练，可教授不同课程，但教练并非全程有空。
In addition, trainers are not skilled in all types of classes.
此外，教练并非精通所有课程类型。
Certain trainers can only teach certain types of classes.
部分教练仅能教授特定课程。
Therefore, he needs a system that will allow him plan classes based on the trainer availability.
因此，他需要一套系统，可根据教练空闲时间规划课程。

Project Requirement

项目要求
After several rounds of discussion, Mr Arnold would like to have the following essential features to be included.
经多轮沟通，阿诺德先生要求系统包含以下核心功能：

- Customer system that allows customers to sign up for fitness package, and register for classes.
  客户系统：支持客户购买健身套餐、预约课程。
- Admin system that allows the admin to register staff schedules, register class schedules, check customer info, and view class statistics and monthly statistics.
  管理员系统：支持管理员录入教练日程、设置课程表、查询客户信息、查看课程统计与月度统计。

Main Menu

主菜单
The application should display the main menu when the user first opens the application
用户打开程序时，需显示主菜单：

```
@@@@ Fitness Management Portal @@@@
1. Customer registration/Login
2. Admin Login
3. Exit
Enter option:
```

```
@@@@ 健身管理门户 @@@@
1. 客户注册/登录
2. 管理员登录
3. 退出
请输入选项：
```

Customer system

客户系统
The customer screen will show 2 options: Login or Register.
客户界面显示两个选项：登录或注册。
For registration, the customer will need to enter:
注册时客户需填写：

- Username (must be unique)
  用户名（必须唯一）
- Password
  密码
- Email address
  电子邮箱
- Age
  年龄
- Gender
  性别

After registration, the customer can login using the username and password.
注册完成后，客户可使用用户名和密码登录。
After login they will have the following options:
登录后显示以下菜单：

```
@@@@Members Main Menu@@@@
1. Buy package
2. View packages
3. Register for class
4. Logout
```

```
@@@@会员主菜单@@@@
1. 购买套餐
2. 查看套餐
3. 预约课程
4. 退出登录
```

Buy Package

购买套餐
To keep the choices simple, Mr Arnold only allows 1 package to be used for 1 type of class.
为简化选择，阿诺德先生规定**一个套餐仅对应一类课程。**
The packages are:
套餐如下：

1. Pilates x 10 (300,30/class)
   普拉提10次卡（300新元，30新元/节）
2. Pilates x 30 (660,22/class)
   普拉提30次卡（660新元，22新元/节）
3. Yoga x 10 (300,30/class)
   瑜伽10次卡（300新元，30新元/节）
4. Yoga x 30 (660,22/class)
   瑜伽30次卡（660新元，22新元/节）
5. MMA x 10 (450,45/class)
   综合格斗10次卡（450新元，45新元/节）

6. MMA x 30 (1050,35/class)
   综合格斗30次卡（1050新元，35新元/节）
7. Boxing x 10 (450,45/class)
   拳击10次卡（450新元，45新元/节）
8. Boxing x 30 (1050,35/class)
   拳击30次卡（1050新元，35新元/节）
9. KPop Fitness x 10 (250,25/class)
   韩流健身10次卡（250新元，25新元/节）
10. KPop Fitness x 30 (600,20/class)
    韩流健身30次卡（600新元，20新元/节）

Payment is done outside the system, so you can automatically add the package once the customer selects the package that they want.
支付在系统外完成，客户选定套餐后，系统自动为其添加套餐权益。

View Package

查看套餐
Customers can see the number of lessons they have left for each package.
客户可查看各套餐剩余课程次数。
Once it reaches zero the item is removed.
次数为0时，该套餐自动移除。

Registering for class

预约课程
Customers can register for classes for the current day, and up to 1 month in advance.
客户可预约当日及未来最长1个月内的课程。
In this screen, it should immediately display the current week classes from Mon to Sunday.
该界面直接显示当周周一至周日的课程，示例如下：

```
Mon (02-Mar)
1. MMA, 8am - 9am, John Wick, Studio 3, 4 slots left
2. Boxing 8am – 9am, Rocky Tan, Studio 4, 7 slots left
3. Pilates 9am – 10am, Linda Lee, Studio 1, 11 slots left
...
```

```
周一（3月2日）
1. 综合格斗 8:00-9:00，约翰·威克，3号教室，剩余4个名额
2. 拳击 8:00-9:00，洛基·陈，4号教室，剩余7个名额
3. 普拉提 9:00-10:00，琳达·李，1号教室，剩余11个名额
...
```

Choose the class number you want to register for or “N” for next week, or “P” for previous week.
输入课程编号预约，输入**N**查看下周，输入**P**查看上周。

The details shown for every class is the class type, start time and end time, the trainer’s name, the Studio name, and the number of slots left.
每节课程显示：课程类型、起止时间、教练姓名、教室名称、剩余名额。
The number in front is a running number, to make it easier for the user to enter their choice.
前方数字为连续编号，方便用户选择。

If the week is the current week, the user should not be able to go back to previous week, and he cannot book lessons that are already past.
当周不可查看上周，且不可预约已过期课程。
The user can only go forward 4 weeks in advance.
最多可提前预约**4周**内的课程。

After entering the number, the user will be asked to confirm the booking.
输入编号后，系统提示确认预约：

```
Confirm booking 03 Mar Pilates 8am – 9am? (y/n): y
Your package has been deducted
```

```
确认预约3月3日普拉提8:00-9:00？（是/否）：是
套餐次数已扣除
```

If the user types “y” to confirm, their package is automatically deducted.
输入**y**确认后，系统自动扣除套餐次数。
If they do not have the package, the system should not allow them to register for that lesson.
无对应套餐时，系统禁止预约该课程。

Admin System

管理员系统
After the login, the admin should see the following menu:
管理员登录后显示以下菜单：

```
@@@Admin System@@@
1. Class planner
2. View/Add trainer
3. View statistics
```

```
@@@管理员系统@@@
1. 课程规划
2. 查看/添加教练
3. 查看统计数据
```

Class planner

课程规划
The admin can set the day, and view the classes scheduled for that day.
管理员可选择日期，查看当日课程安排。
The admin will have the option to delete class or add a new class.
管理员可删除或新增课程。
Deleting a class should not be allowed if students are already registered in that class.
已有学员预约的课程**不可删除。**

When adding a class, the admin must enter the following:
新增课程需填写以下信息：

- Class type (Yoga, Pilates, MMA, Boxing, KPop Fitness)
  课程类型（瑜伽、普拉提、综合格斗、拳击、韩流健身）
- Start time (earliest start time is 8am and latest start time is 8pm. All classes are 1 hour)
  开始时间（最早8:00，最晚20:00，所有课程时长1小时）
- Studio name (There are 4 studios. Studio 1 and 2 can hold 20 students, Studio 3 and 4 can hold 10 students)
  教室名称（共4间：1、2号大教室容纳20人，3、4号小教室容纳10人）
- Trainer name
  教练姓名

The system must do the relevant error checking, including:
系统需进行以下校验：

- Class type is in 1 of the 5 categories
  课程类型为指定5类之一
- Start time cannot be before 8am and after 8pm.
  开始时间在8:00-20:00之间
- There cannot be another class in the studio at that time. This includes overlapping time, e.g. 1 class at 3pm and another class at 3.30pm
  同一教室同一时间段无冲突课程（含时间重叠，如15:00与15:30课程）
- Trainer must be able to teach that class and must be available
  教练可教授该课程且时段空闲

View/Add trainer

查看/添加教练
This section allows the admin to see the list of available trainers, their class type, and their hourly rate.
该模块支持管理员查看教练列表、可授课程及时薪，示例如下：

```
Trainer list
1. John Wick (MMA, Boxing) $120/hr
2. Rocky Tan (MMA), $100/hr
3. Linda Lee (Pilates, Yoga), $85/hr
...
Enter trainer number to view schedule, ‘A’ to add new trainer, 0 to go back:
```

```
教练列表
1. 约翰·威克（综合格斗、拳击）120新元/小时
2. 洛基·陈（综合格斗）100新元/小时
3. 琳达·李（普拉提、瑜伽）85新元/小时
...
输入教练编号查看日程，A添加新教练，0返回：
```

Trainer schedule

教练日程
Trainers' schedules are the same every week.
教练日程**每周固定。**
This means the admin only has to set the schedule from Mon to Sunday.
管理员仅需设置周一至周日的日程，每周循环。
Appendix A has a list of trainers and their available timeslots in a text file.
附录A为教练及空闲时段文本文件，作为程序启动默认数据。
This data should be used as the default data when the program starts.
程序启动时加载该默认数据。
However, the admin should still be able to change the trainers and schedules.
管理员可修改教练信息与日程。

Statistics

统计数据
Statistics is a key feature that will allow Mr Arnold to run his business better.
统计功能是助力阿诺德先生优化运营的核心模块。
Some statistics required are:
需包含以下统计项：

1. Class registrations by class type
   按课程类型统计预约人数
2. Packages bought by class type
   按课程类型统计套餐销量
3. Top 5 trainers
   人气TOP5教练
4. Profits earned*
   盈利计算*

All statistics must allow display by last 4 weeks, last 3 months or last 12 months.
所有统计支持查看**近4周、近3个月、近12个月**数据。

*The rental for the 4 studios are fixed at
$17000 permonth. *4$间教室月固定租金17000新元。$Fora1hourclass, theadditionalelectricityandcleaningcostis$20 for big studio, and
$12 forsmallstudio.$ 每节1小时课程，大教室额外水电清洁费20新元，小教室12新元。$Tosimplifytheprofitcalculation, theamountearnedperstudentisseta$
35 per class.
例：综合格斗单节课收入35新元/人。

For meaningful statistics, you must have at least 300 students and 1500 class data over 6 months in your submission.
提交时需包含**至少300名学员、6个月内1500条课程数据**，保证统计有效性。

Additional marks will be awarded for graphical display of statistics and any additional statistics shown.
使用图表展示统计数据、新增统计项可获得附加分。

Data Management

数据管理
For this project, you may choose any of the following options to store your data:
本项目可选用以下方式存储数据：

- MySQL or SQLite
  MySQL或SQLite数据库
- csv or txt format
  csv或txt文本格式
- json format
  json格式

All datafiles must be submitted together with Python files during submission.
所有数据文件需与Python代码一并提交。

Instructions for CA3 Report, Python Codes and User Guides

CA3报告、Python代码及用户指南说明

Cover Page

封面
The cover page should include the institution name (and institution logo) the programme and the module name, the semester and year and date of submission.
封面需包含：院校名称（及校徽）、专业、课程名称、学期年份、提交日期。

All these must be centralised in the page.
所有内容居中排版。

Write FULL Name and Student number as in the register on the cover.
封面填写学籍登记的**全名**与**学号**。
Students should a keep a copy of assignment submitted.
学生需保留作业提交副本。

Python Codes

Python代码
Suggested Python version: Python 3.8.2 and above
推荐Python版本：3.8.2及以上

Please zip all your Python codes into single file and upload it together with the report.
将所有Python代码压缩为单个文件，与报告一并上传。
If you used any additional Python library apart from the standard package, you need to include them in the submission.
若使用标准库以外的第三方库，需一并提交。

Referencing

引用
No referencing is needed for program designs and codes.
程序设计与代码无需引用文献。

Font and Spacing

字体与行距
Font: Times New Roman
字体：Times New Roman
Font size: 12 and 1.5 for line spacing.
字号：12号，行距1.5倍

Penalty Marks for Late Submission of Assignment

作业迟交处罚
By one day: 20% to be deducted from total marks.
迟交1天：扣总分20%
More than one day: submission will NOT be graded.
迟交超过1天：作业不予评分

Use of A.I Tools

AI工具使用
As the industry moves towards using A.I for coding, the school is also moving ahead to prepare students for this change.
随着行业普及AI编码，学院同步培养学生适配该趋势。
However, the understanding of the code and testing of the code is still essential.
但学生必须理解代码并完成测试。
Students must be able to understand every aspect of their code and any errors or flaws in the code are the sole responsibility of the student.
学生需完全理解代码，代码错误与漏洞由学生自行负责。

Plagiarism and Collusion

抄袭与串通
Students are not allowed to reuse old assignments or submit projects from previous semesters or copy largely from sources, particularly from the Internet web.
禁止复用旧作业、提交往届项目或大量抄袭来源内容（尤其网络资源）。

The submitted report must show evidence that this is students' own work.
提交报告需证明为学生原创。
No marks will be awarded if there are no workings or reasonable explanations.
无操作过程或合理解释的作业不得分。
Please be reminded that plagiarism and collusion is a serious offence, and all cases will be referred to the administration.
抄袭与串通属于严重违纪，所有案例将上报校方。
Grades will be withheld if the submission is suspected of plagiarism or collusion till investigations are completed.
涉嫌抄袭/串通的作业暂停评分，直至调查结束。

Important Dates of CA2

CA2重要日期
CA2 Group Assignment Deadline: 11 May 2026 11.59 a.m.
CA2小组作业截止时间：2026年5月11日上午11:59

Submit your project via Canvas, submission must be completed for reports to be graded.
通过Canvas提交项目，完整提交后方可评分。

Lecturer Contact

讲师联系方式
You should contact your lecturer via your SIM email whenever you have any issue about your project.
项目相关问题请通过学院邮箱联系讲师。

Marking Rubric

评分标准
（因表格篇幅较长，核心内容中英对照如下）

1. **项目报告（10分）**：程序描述、流程图、展示清晰度、创意性
2. **Python代码（50分）**：项目理解、模块应用、算法逻辑、附加功能、用户界面
3. **用户指南（15分）**：截图示例、测试数据完整性
4. **口头汇报（25分）**：表达清晰度、演讲风格、内容匹配度、问答环节、准备情况
5. **非AI模块演示（25分）**：技术难度、代码理解、展示清晰度